

HASSAN YOSUF

Backend Engineer - Java & AI

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PROFESSIONAL SUMMARY

Backend Engineer with ~4 years of experience designing scalable, production-grade distributed systems. Strong expertise in microservices architecture, REST APIs, event-driven systems (Apache Kafka), and distributed caching (Redis). Cloud-agnostic developer certified in AWS, GCP, and Snowflake, with a strong focus on the full SDLC - CI/CD automation, unit testing, and performance optimisation. Delivered high-throughput data pipelines processing 200K+ records in under 15 minutes. Actively experimenting with RAG workflows and LLM APIs (Claude, LLaMA) to build AI-augmented backend solutions.

TECHNICAL SKILLS

Frameworks and Tools: Spring Boot, Spring Security, Spring Data JPA/Hibernate, Microservices Architecture, Apache Kafka, Redis, Node.js, J2EE, REST APIs, Maven, API Gateway. Multithreading, Concurrent Batch Processing

Cloud and DevOps: AWS (Certified), GCP (Certified), Snowflake (Certified), Docker, Git, CI/CD Pipeline (Jenkins), Linux, Kubernetes

Databases: MySQL, PostgreSQL, Oracle, Snowflake, MongoDB, Cassandra (familiar)

GenAI & LLM: RAG Workflows, LLM APIs (Anthropic Claude API, LLaMA), Prompt Engineering, AI Agents (exploring), OpenText

Software Engineering Concepts: DSA, MVC Architecture, OOPS, Design Patterns, API Design, SDLC, Agile Methodology, TDD (JUnit), JWT, OAuth2

Other Tools: Claude, Cursor, Tableau, Power BI, JIRA, Confluence

PROFESSIONAL EXPERIENCE (3.9+ YEARS)

Backend/ Framework Engineer

Quantiphi Analytics Solution Pvt. Ltd. — Bengaluru, India

December 2022 – Present

- **Real Estate Tokenization Platform:** Designed and developed a cloud-based real estate tokenization platform on GCP, enabling fractional property investment through Bitcoin. Worked on core microservices – User Service (auth & user management), Asset Service (token valuation & tracking), Ternas Service (property listing), and Finance Service (payout logic) - deployed on GCP Compute Engine for scalability and high availability.
- **Microservices Architecture (Java):** Architected and maintained robust Java Spring Boot microservices using REST APIs for synchronous service communication and Apache Kafka for asynchronous event-driven messaging – applying design patterns to decouple services and enable independent scaling of financial transaction modules across user lifecycle, asset valuation, and payout workflows.
- **Secure Bitcoin & Fintech Transactions:** Built APIs for token purchase, property allocation, and profit distribution; integrated Fireblocks API for secure Bitcoin transactions. Designed Finance Service modules with batch/cron jobs for periodic profit-sharing distribution based on token ownership share.
- **Cloud Data Contribution (GCP):** Worked on data persistence using Cloud SQL for relational data and GCS for property documents and transaction logs; supported BigQuery integration for analytical queries – contributing to transparent tokenization and transaction handling at scale.
- **Scalable Data Engineering (Spark/Snowflake):** Designed and developed production-grade data processing modules using Spark and Snowflake, optimising ingestion logic from 5+ distributed sources to process 200,000+ records in under 15 minutes with high availability.
- **GenAI & LLM Integration:** Built RAG-based experiments integrating LLMs APIs – exploring retrieval-augmented generation pipelines, prompt engineering, and LLM-powered workflow automation for internal tooling use cases.
- **AI Service Integration:** Integrated AI-powered speech-to-text and Google DLP APIs to automate PII redaction workflows in production - directly embedding AI services into backend delivery pipelines and reducing manual compliance overhead.
- **CI/CD & Production Operations:** Streamlined software delivery via CI/CD pipelines and Git branching strategies; conducted root cause analysis for production issues and implemented Redis-based distributed caching for high-frequency data access, centralised logging, and health checks to ensure system reliability.
- **Cross-functional Collaboration:** Partnered with stakeholders to gather requirements, design workflows, and align business logic with technical implementation in a fast-paced Agile environment.

Tech Stack: Java, Spring Boot, Spring Security, JWT, Apache Kafka, Redis, Python, Spark, Snowflake, GCP, Docker, AWS, CI/CD, Git, REST APIs, PostgreSQL, MySQL, Fireblocks

PROJECTS

ChatUp - Real-Time Messaging Backend

- Built a real-time messaging backend using Spring Boot with WebSocket/STOMP, enabling low-latency bi-directional communication and real-time state synchronisation via an event-driven architecture.
- Integrated an AI chat assistant using the Groq API (LLaMA 3.3), intercepting *@ai-prefixed* WebSocket messages, processing them through a Spring Boot service, and broadcasting AI-generated replies in real time to all room subscribers.
- Developed a responsive frontend using HTML5, CSS3, and ES6+ JavaScript with SockJS for WebSocket fallback, cross-browser compatibility, and distinct UI rendering for AI vs human messages.
- Dockerized the application and deployed on Render for consistent, scalable hosting.

Tech Stack: Java, React, Spring Boot, JavaScript, WebSocket, STOMP, SockJS, Docker, Render, Groq API, LLaMA 3.3, LLM Integration, MVC Architecture

CERTIFICATIONS

AWS Certified Cloud Practitioner | Google Cloud Platform – Associate Cloud Engineer | Snowflake SnowPro Core Certification

EDUCATION

Bachelor of Technology – Computer Science & Engineering

Integral University, Lucknow, India

August 2018 – June 2022